

### The Utility Problem

Almost every utility is involved in reducing non-technical losses - more or less successfully. However, it is very difficult to find the ideal way to achieve the best results. Some utilities are just starting loss reduction projects and if the utility does not already have procedures in place to reduce non-technical losses, it can be very difficult to deliver new procedures without prior experience in this area. Developing a methodology can take years, during which many problems can arise, but these can be eliminated by implementing already established and tested methods.

In creating a methodology for an effective loss reduction programme, it is necessary to implement a system that should perform multiple functions related to the management and performance of non-technical loss investigations. These functions are primarily:

- Planning - it is primarily a systematic prioritization of whether to investigate a given area or location. Decisions can be based on a “tip” or can be based on a risk analysis where possible risks of non-technical losses are calculated and evaluated.
- Analysis and evaluation - For the successful and correct conclusion of the investigations, it is necessary to have tools for proper investigation analysis and evaluation. For that, it is necessary to see the data in context (the correct interpretation of the data in space, in context with temperature, other grid elements, and time series). This increases the possibilities of exploring and testing the results for the correct identification of non-technical losses.
- Case management - is important because of the systematic approach when it is necessary to investigate an area divided into smaller parts. Case management also allows multiple activities to be combined into a single unit, which is beneficial for evaluation and reporting on non-technical losses.
- Information exchange and audit trail - investigators should be able to add their thoughts and other information to the investigations. These are often added as images or comments.
- Archiving & record keeping - are important, especially for providing the information in court hearings.
- Reporting - reporting enables utilities to compare the performance of the non-technical loss reduction program with expectations and other programs.

Reducing losses is very important for the company's profits. Losses have to be purchased at a certain price or market price and therefore, the company's profits are reduced. . If losses are reduced, the company's profits increase and saved resources can be invested elsewhere.

### The Data Solutions and Methods

Awesense’s Energy Transition Platform provides end-to-end non-technical loss investigation management from planning to archiving and reporting. To address the areas mentioned above, the Awesense Platform uses the following tools:

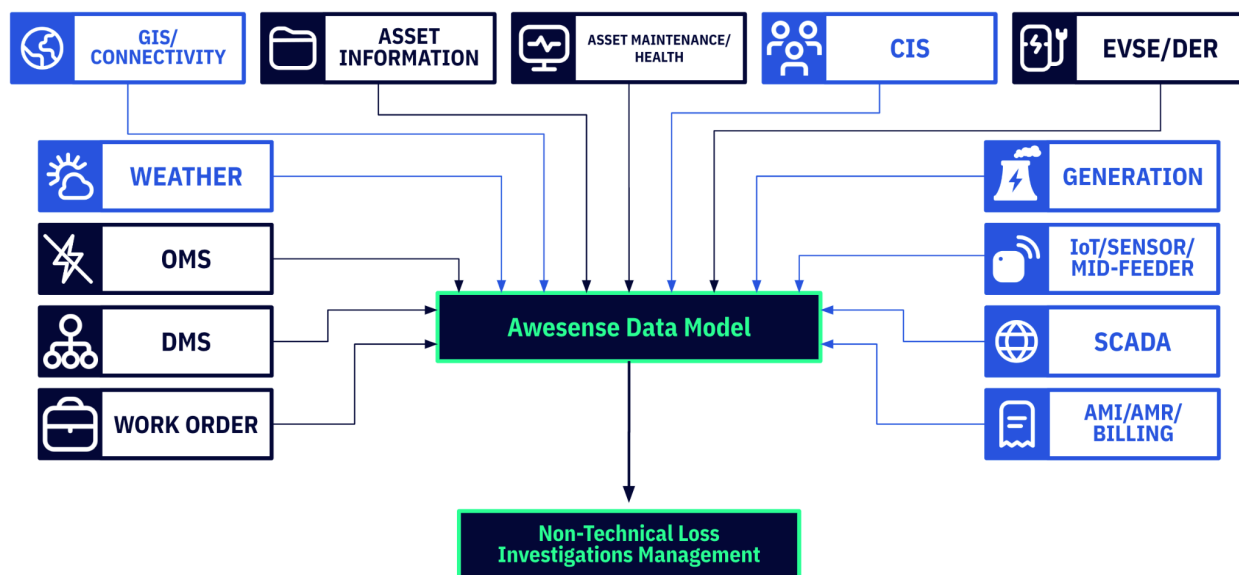
- Planning:
  - This step can be approached in several ways. One is on the “tip” basis, where areas or sites are identified for closer investigation based on the technicians' local knowledge or denunciation, etc. This method is rather complementary to the Awesense Platform's main method, which is identification by risk analysis. This method uses Awesense’s Segment Analyzer, where an analysis model containing individual factors is applied. The analysis factors can be defined either based on internal grid records or also on publicly available data from various trusted public sources - for example, billing irregularities, higher crime rates, customer demographics, unusual consumption patterns, type of connection, etc. The Segment Analysis then applies the selected model to a group of segments and ranks the segments according to the selected factors. This simplifies the decision-making process on which segment to perform appropriate investigations to detect and reduce losses.
- Analysis and evaluation:
  - The energy balance can be displayed directly in Awesense’s True Grid Intelligence (TGI) frontend solution. In detail, it is possible to display the energy balance graph with different display options, calculate the balance for kWh or Ah, display the balance for individual phases or for all phases, choose a bar or line graph, add graphical display of weekends and nights, add temperature display, etc. The investigation sites can be seen on a map with different satellite tiles or street views. All this helps to add context and makes the energy balance conclusion easier to evaluate.
- Case management:
  - Based on the segment analysis and the investigation of the energy balances, suitable segments are selected for the field investigation. Individual investigations then identify locations where unauthorised consumption or other losses are suspected and help to identify their source. The methodology is based on a gradual reduction of the selected segments (drill-down approach). The size of the segments and investigation areas vary from project to project and from utility to utility. The baseline investigation is performed on large segments with several hundred customers. Exploratory investigations are then performed on sub-segments of the baseline investigation. Detailed investigations then investigate the segments whose results from the previous investigation show a positive result in the form of probable losses in the grid and then move on to the so-called specific investigation, which essentially looks for the source of the cause of the losses at the specific consumption point.
- Information exchange and audit trail:
  - Awesense Platform’s investigation instance can include additional data in the form of comments or images. This is especially important for gathering evidence about unlawful conduct. To stay organized, the investigations also have an audit trail to observe any change or modification to the investigations.
- Archiving & record keeping:

- All investigations can be created and the results can be viewed directly in TGI. All findings can be documented and therefore, evidence can be gathered based on these evaluated investigations for further use. Everything is very simple and intuitive. All records can be exported to PDF and shared with anybody.
- Reporting:
  - Awesense Platform offers multiple ways of reporting. TGI frontend offers no-code reporting and dashboarding capabilities. For more complex reporting, the set of APIs is available for connection of the preferred reporting tool like BI tools or spreadsheets.

Thanks to Awesense's previous experience with such projects, the implementation of these processes is already almost seamless and very fast. Awesense can also bring a new perspective to an already established methodology in the utility.

### Required Data Sets

The required data for this use case are visualized in the following diagram. Data sources highlighted in [royal blue](#) are required for this use case.



### Available Use Case Implementations:

Users of the Awesense Energy Transition Platform can access this use case by the following means:

1. TGI Investigations Management - Please speak to Awesense about access to the sandbox environment or if you would like to see a demo of the solution.